Two-page summary

2026-11-09 17:58 PST

Toward an optimal data-generating process for simulating biomarker-treatment interactions in aggregated N-of-1 trials (pmsimstats team).

Goal

A predictive biomarker is validated by an interaction test. With two equal subgroups and an interaction as large as the main effect, that test needs about four times the sample of a main-effect test (Brookes et al. 2004). Aggregated N-of-1 designs have been proposed to make it affordable. Their power is estimated by simulation, and a simulation can only show what its data-generating process (DGP) allows. Paper 01 asks what that DGP should be.

Approach

We start from the one DGP we found that encodes a predictive biomarker, the multivariate normal model of Hendrickson et al. (2020). We call it the starting DGP. In it the biomarker acts as a correlation with the drug response. We take the elements of the DGP one at a time, compare the candidate choices for each, and judge them by five criteria:

1. existence (the joint distribution must be valid);
2. identifiability under carryover;
3. fidelity to the pharmacology;
4. analyzability;
5. calibration status.

The choices are collected as the recommended DGP, a step toward an optimal one. We derive the closed-form power of a paired-difference interaction statistic (E9) and a Schur-complement bound on the largest feasible interaction. We check both by Monte Carlo simulation. Three simulation studies verify the closed forms, rerun the starting code, and stress-test ten analyses under a prognostic biomarker. The Hybrid design is the primary case, and all simulations use the starting response and baseline parameters.

Results

Feasibility. On the grid of the earlier study, the starting code repaired 20 correlation matrices (25%) that were not positive definite, without a warning. The recommended DGP refuses such a matrix instead. On the extended grid it finds 75% of cells infeasible at an interaction correlation of $c_{bm} = 0.6$. This choice rests on the existence criterion.

The biomarker-response correlation under carryover. This is the element that matters most. Under the constant rule, the correlation stays at its on-drug value after first exposure. The interaction is then absent among the post-baseline visits, and power falls nearly to the test size at every positive half-life. Under the decaying rule, the correlation decays at the carryover half-life. Power then stays near its no-carryover value at pharmacological half-lives of 0.1 to 0.2 week and declines smoothly at longer ones. Power depends on the rule only through the on/off correlation difference. The starting code shows the same pattern. At 0.1 week its power was 0.09 with the constant correlation and 0.64 with the decaying one, against 0.67 without carryover. The choice of the decaying rule rests on identifiability under carryover.

Carryover on the mean. The starting code compounds the decay of the drug-response mean across off-drug visits. The recommended DGP anchors it at the last on-drug mean. The choice rests on fidelity to the pharmacology, as a judgment; its effect on power is at most 0.006 either way.

Serial correlation. The correlation structure bounds the largest interaction the DGP can hold: 0.256 under compound symmetry and 0.474 under separable AR(1) at a serial correlation of $\rho = 0.7$. Across ρ from 0.4 to 0.95 the bound is largest at 0.7 and falls steeply above it. The value 0.7 is a convention. A PTSD symptom series suggests 0.9, where the bound falls to 0.34 and the Hybrid design needs 154 participants at $c_{bm} = 0.25$ instead of 278.

Prognostic biomarker. If a prognostic biomarker acts through a pathway absent at baseline, analyses that share its effect with baseline are anticonservative. A prognostic correlation also shares the biomarker row with c_{bm}, so each lowers the feasible room for the other.

Planning. We give the total sample size for 80% power of E9 and the simulated power of the recommended analysis G10 (a random-intercept CAR(1) mixed model with an analysis half-life chosen by AIC on the model without the interaction, and CR2 standard errors). G10 rejected the null at 0.046 to 0.053. At the E9 sample size, G10 power was 0.844 to 0.936 in the 20 uncapped cells, so the closed-form rule is conservative for G10. In the Hybrid design the G10 sample size for 80% power is 0.71 to 0.83 times the E9 value. At $\rho = 0.9$, 8 to 12% of G10 fits failed in each crossover cell at $N = 70$. The fitted serial correlation reached its limit of 1 and could not be told apart from the random intercept.

The recommended DGP

The recommended DGP has a biomarker-response correlation that decays after discontinuation, carryover on the mean anchored at the last on-drug value, a separable AR(1) structure at $\rho = 0.7$ with a cross-component correlation of 0.2, and a check that refuses infeasible effect sizes. Path allocation is unchanged from the starting DGP. Every parameter is labeled as a starting value, a choice or uncalibrated.

Conclusion

Three points are to be emphasized. First, the biomarker-response correlation must decay after discontinuation; a constant correlation removes the interaction under any carryover. Second, the serial correlation structure sets the largest interaction the DGP can hold, and the required sample size depends strongly on ρ above 0.7. Third, a matrix outside the feasibility bound should be refused rather than repaired. We propose the recommended DGP as a reference for simulation studies of these designs. Calibrating the serial correlation, the drug-response SD and the symptom half-life is the next step toward an optimal DGP.